

Analytics for a Better World

Predicting while optimizing: mobile clinic routing with learned constraints

Joaquim Gromicho

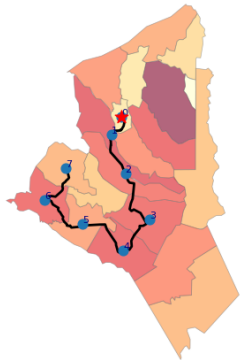
joint work with Mayukh Ghosh and Chintan Amrit

Amsterdam Business School, University of Amsterdam

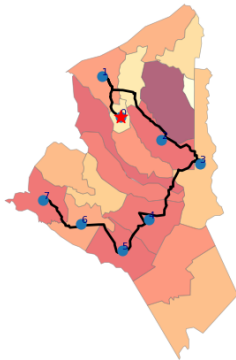
Chapter 6 of Mayukh Ghosh's PhD thesis, *Domain Driven Analytics*

Same history, same solver, same number of stops

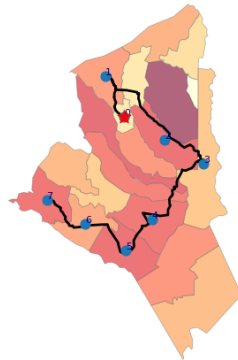
cannibalisation: static (I2)
8 stops, 56 km, promised 1892, realised 1374



cannibalisation: learned MLP (I3)
8 stops, 79 km, promised 1664, realised 1621



cannibalisation: oracle
8 stops, 79 km, promised 1653, realised 1630



Left: 1892 doses promised, 1374 delivered.

Middle: 1664 promised, 1621 delivered.

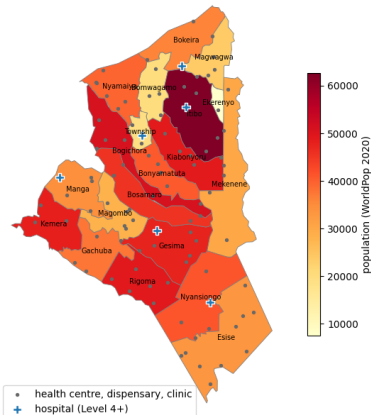
Why?

The setting: one mobile clinic, many wards

- AMREF Health Africa ran mobile clinics for COVID-19 vaccination in Kenya (Ghosh, Amrit, Gromicho; thesis Chapter 6).
- One clinic, a planning period of a few weeks, at most a handful of stops.
- Vaccines are collected at a referral hospital, so the route is an open path from town.
- Which wards to visit, and in what order?

Today: Nyamira County, 20 wards, 745 thousand people, 897 km².

Nyamira County: wards, population and health facilities



Predict-then-optimize, and its hidden assumption

- 1 Predict demand $\hat{d}_i$ per ward from population, unvaccinated share, accessibility.
- 2 Solve a prize-collecting routing problem that rewards $\hat{d}_i$ at visited wards.

Hidden assumption

Demand at ward i does not depend on which other wards the clinic visits.

Two reasons it does:

- **Cannibalisation:** people near a visited ward get their dose there; a later visit next door finds less demand.
- **Mobilisation:** health volunteers, market days and word of mouth raise turnout around a visited ward.

Spillover in one formula

Let $y_j = 1$ if the clinic visits ward j , and d_{ij} the road distance.

$$E_i(y) = \min\left(1, \sum_{j \neq i} e^{-d_{ij}/R} y_j\right) \qquad D_i(y) = b_i(1 + \gamma E_i(y)) \varepsilon_i$$

- E_i : exposure of ward i to visits nearby, range R (10 km today); one adjacent visit gives about 0.6, two saturate it.
- b_i : base demand if visited alone; grows with population, unvaccinated share, distance to the nearest facility.
- $\gamma < 0$ cannibalisation, $\gamma > 0$ mobilisation, $\gamma = 0$ the textbook case.
- The clinic administers at most 250 doses per visit.

Real turnout data would replace the simulator; the structure of the question stays.

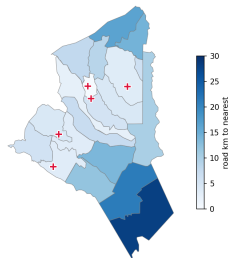
Open data, no API keys

Ingredient	Source
Ward boundaries	GADM 4.1, level 3
Population per ward	WorldPop 2020, 1 km, UN-adjusted
Road network	OpenStreetMap through osmnx
Health facilities, geocoded	Maina et al. (2019), <i>Scientific Data</i> 6:134, CC0
Road distances between wards	pandana contraction hierarchies, 20 by 20 in milliseconds
Demand	simulator above, both signs of γ
Optimization	Gurobi, under the free Community Edition limit

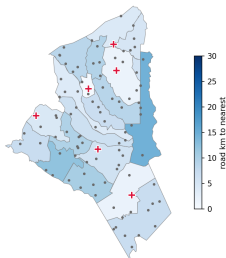
Nyamira has 98 geocoded public facilities in this list. Only its 6 hospitals (Level 4 and above) store vaccines, so accessibility is the road distance to the nearest of those.

Same count, different facilities: why the source matters

OpenStreetMap: 6 tagged facilities



Maina et al. 2019: 6 hospitals of 98 facilities



The conclusion survives the change of source; the route and the wards it favours do not. Maina et al. is public sector only, from about 2018.

	OSM	Maina et al.
Facilities in Nyamira	6	98
Used as vaccine points	6	6
Of those, real hospitals	2	6
Distinct sites	5	6
Mean road km to nearest	9.0	6.8
Worst ward, road km	28.2	14.6
Static: promised, delivered	1889, 1331	1892, 1374
Learned: promised, delivered	1551, 1530	1664, 1621

- OSM tags two dispensaries and a health centre as hospitals and lists one site twice.
- Only one site is common to both: the county hospital in Nyamira town.
- Ward accessibility rankings are unrelated, rank correlation 0.06.

Learning a demand model that contains the decision

Where does a training set with y in it come from? Replay the history of past deployments.

period, ward	log pop	unvaccinated	inaccessibility	exposure E_i	turnout D_i
1, Esise	10.4	0.41	1.00	0.31	361
1, Gesima	10.8	0.52	0.49	1.12	275
2, Gesima	10.8	0.52	0.49	0.08	560
$\vdots$					

- 1500 random deployments of 3 to 10 wards, 30 000 rows.
- Two models: linear regression (transparent, mis-specified) and an 8-unit ReLU network, fitted with L-BFGS in seconds.
- In practice: past periods' routes plus ticketed turnout, the feedback loop AMREF asked for.

The routing model

Wards N , depot s , arcs A , road distance c_{ij} , cost μ per km, stop limit K , capacity C per visit, spillover weights w_{ij} , ward features f_i , $M = 4C$. $x_{ij}, y_i \in \{0, 1\}$: drive $i \rightarrow j$, visit i ; $E_i \in [0, 1]$ exposure; $\hat{d}_i \geq 0$ modelled demand; $z_i \in [0, C]$ doses delivered.

$$\begin{aligned}
 \max \quad & \sum_{i \in N} z_i - \mu \sum_{(i,j) \in A} c_{ij} x_{ij} \\
 \text{s.t.} \quad & y_s = 1, \sum_j x_{sj} = 1, \sum_j x_{js} = 0 && \text{path starts at the depot} \\
 & \sum_{j \neq i} x_{ji} = y_i && \forall i \neq s \\
 & \sum_{j \neq i} x_{ij} \leq y_i && \forall i \in N \\
 & \sum_{(i,j) \in A} x_{ij} = \sum_i y_i - 1, \quad \sum_i y_i \leq K \\
 & E_i = \min(1, \sum_{j \neq i} w_{ij} y_j) && \forall i \in N \\
 & \hat{d}_i \leq g(f_i, E_i) + M(1 - y_i) && \forall i \in N \quad \text{the demand model} \\
 & z_i \leq \hat{d}_i, \quad z_i \leq C y_i && \forall i \in N \\
 & \sum_{i,j \in S} x_{ij} \leq \sum_{i \in S} y_i - y_k && \forall S \subseteq N \setminus \{s\}, k \in S \quad \text{DFJ, lazy}
 \end{aligned}$$

The DFJ family is generated in a MIPSOL callback with cbLazy: follow the successors from the depot, every visited ward not reached lies on a cycle S . Details and the MTZ comparison: MO-book, Traveling Salesman Problem.

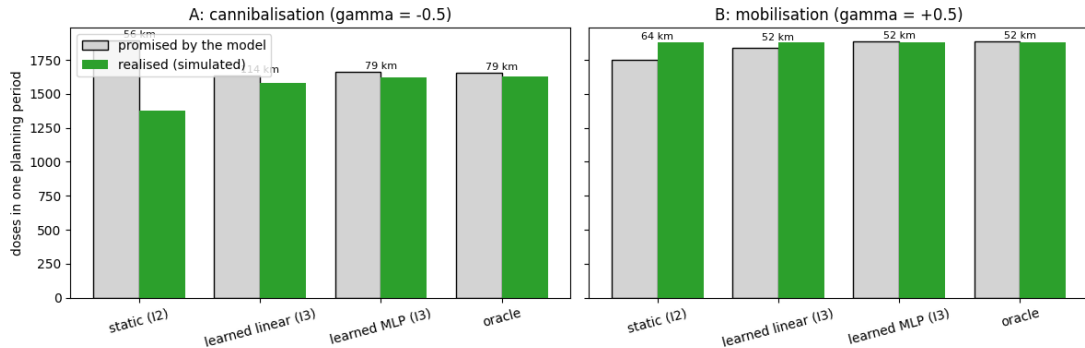
Three ways to fill in g inside the same routing model

Same path constraints, same objective; only the demand model g changes.

Variant	Thesis analogue	The demand model g
static	Intervention I2	$g(f_i, 0)$, a number fixed before solving
learned	Intervention I3	$g(f_i, E_i)$, the fitted model embedded as constraints
oracle	upper bound	$b_i (1 + \gamma E_i)$, the simulator's own mean

- Embedding: one call, `add_predictor_constr(model, g, features, output)` from `gurobi-machinelearning`.
- Size with the network: 980 variables, 404 constraints, no subtour constraint among them. Solves in about a second.
- The oracle is linear in y once E_i is in the model, so it is an exact MIP, not a simulation.

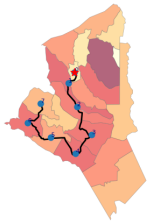
Promised versus delivered



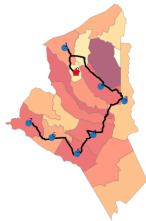
- Cannibalisation: the static plan delivers 73% of what it promises; the learned network delivers 97% and 18% more doses.
- Mobilisation: the same doses, but the static plan drives 64 km where 52 km suffice; doses per km go from 29 to 36.
- The realised-to-promised ratio is a diagnostic any programme can compute after the fact.

The two signs pull the route in opposite directions

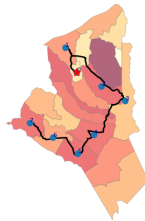
cannibalisation: static (I2)
8 stops, 56 km, promised 1892, realised 1374



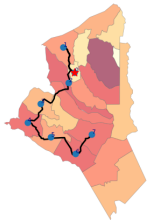
cannibalisation: learned MLP (I3)
8 stops, 79 km, promised 1664, realised 1621



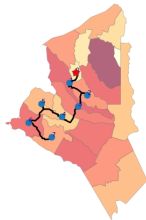
cannibalisation: oracle
8 stops, 79 km, promised 1653, realised 1630



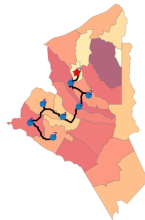
mobilisation: static (I2)
8 stops, 64 km, promised 1750, realised 1882



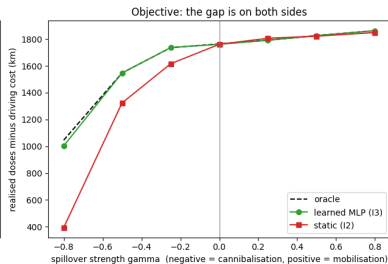
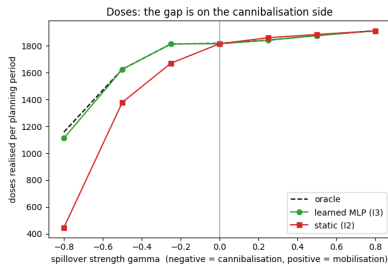
mobilisation: learned MLP (I3)
8 stops, 52 km, promised 1887, realised 1877



mobilisation: oracle
8 stops, 52 km, promised 1884, realised 1879



How much does it matter? Sweep the interaction strength



- At $\gamma = 0$ the three coincide, as they should.
- Cannibalisation costs doses; mobilisation costs kilometres. Only the objective panel shows both.
- Models are retrained per γ ; the learned line tracks the oracle throughout.

Live demo: you predict, I run

The notebook is live in Colab. For each experiment: first your prediction by show of hands, then the rerun.

- ❶ **Range.** R_KM from 10 to 5, then 20. Does the gap between static and learned grow or shrink?
- ❷ **Stops.** K_STOPS from 8 to 4, then 12. At which end can the static planner ignore spillover?
- ❸ **Model size.** HIDDEN from 8 units to 2. What happens to the model size, and to the doses delivered?

Each rerun takes a few seconds: data is cached, every solve is about a second.

Try it yourself, now or later

https://github.com/gromicho/mobile_clinics

Open in Colab from the README. No installation needed.

Debrief and the open question

- The planner's error is structural, not statistical: both planners saw the same data.
- Predicting while optimizing means the prediction is a constraint, not an input.
- Keep the model small enough to embed: a linear or tree model, or a one-layer network.

Open question for research and for practice

In a real deployment, where does the training data with decisions in it come from, and how many periods are needed before the learned constraints beat a careful static forecast?

Notebook and code: `mobile_clinic_routing.ipynb`. Thesis: M. Ghosh, *Domain Driven Analytics*, University of Amsterdam, Chapter 6.